



AR, VR, and MR:

What's Trending?



Augmented reality (AR), virtual reality (VR), and mixed reality (MR) offer immersive and interactive experiences that significantly enhance user engagement and satisfaction. Let's delve into the capabilities of AR, VR, and MR technologies, and their impact on industrial application development.

Technological advancements, particularly in areas like display technology, computer vision, and graphics processing have made augmented reality (AR), virtual reality (VR), and mixed reality (MR) more immersive, interactive, accessible, and affordable. These technologies are being used to improve productivity, training, visualisation, and customer engagement, leading to their adoption in enterprises.

The inconsistent use of terms like AR, VR, and MR can lead to confusion. Different sources define and use these

terms in slightly different ways, adding to the confusion. Understanding how these technologies work and how they differ from each other needs to be addressed. The concepts of augmented, virtual, and mixed reality are illustrated in Figure 1.

Further, Table 1 shows how AR, VR, and MR differ from each other.

Market trends in the AR, VR, and MR ecosystem

AR, VR, and MR are thought to be revolutionary technologies capable of replacing smartphones. It has been a

decade since Google launched Google Glass, which marked the beginning of commercial virtual reality headset development. Several companies such as NVIDIA, Siemens, and Microsoft have recently unveiled their strategies for the industrial metaverse, distinct from the consumer-focused metaverse.

Given the growing demand from end users for advancements in augmented reality technology and the increasing reality of the metaverse, 2024 is poised to be a pivotal year for the future of AR/VR/MR technologies (Figure 2).

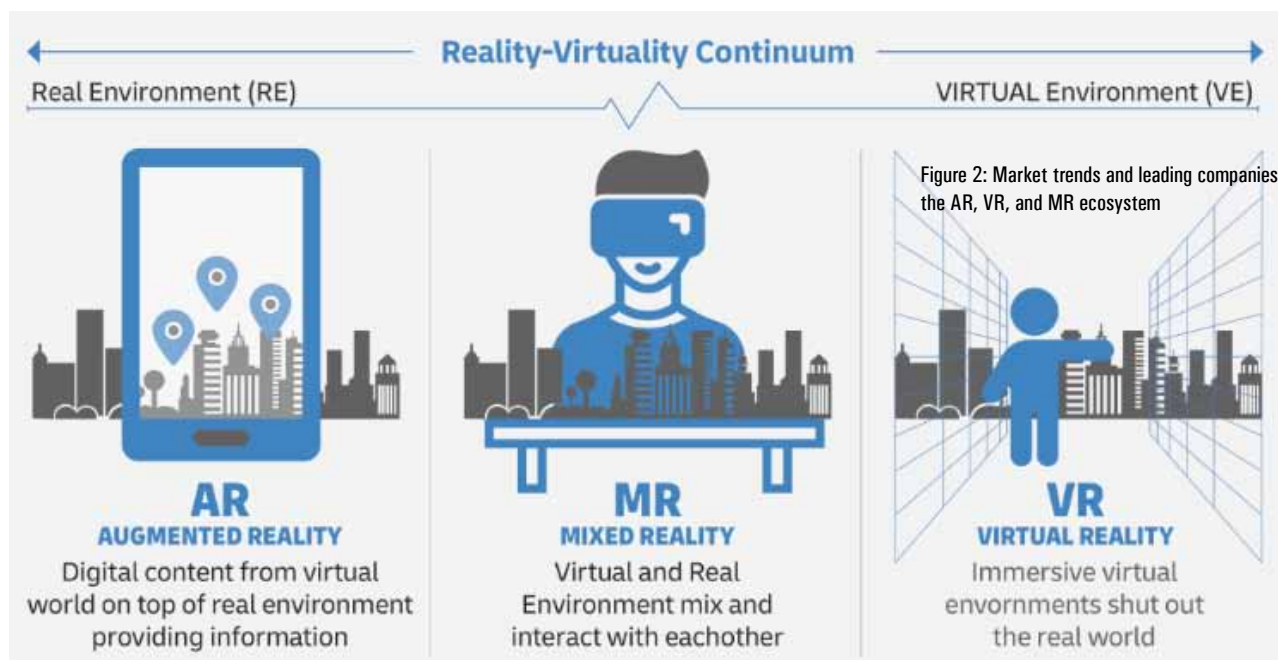


Figure 1: Concept of AR, VR, and MR